

Customer Information

Mr Collins

+233544036297

Eastern Region, Ghana

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Quote #: QT-COLLINS-E43DC5

Date: May 15, 2026

Valid for: 30 days

Off-Grid Hotel Package

Engineered off-grid package for ~332 kWh/day (after diversity) at Koforidua, Eastern Region. 194× 570W panels, 90.0 kW inverter, 330.0 kWh battery storage (24h essential backup basis).

Item	Unit price (GHS)	Qty	Amount (GHS)
Longi 570W Panel	1,500.00	194.0	291,000.00
Energy Precision 10kW 30.0kW Inverter (×3 parallel)	11,000.00	3.0	33,000.00
Energy Precisions 16.0kWh Battery	26,000.00	21.0	546,000.00
Battery cable 35mm copper (per meter)	80.00	20.0	1,600.00
250A MCCB battery breaker	650.00	2.0	1,300.00
Mounting Structure	350.00	1.0	350.00
Mounting rail 18 ft (each)	85.00	90.0	7,650.00
Balance of System (BOS) - 12.0% of equipment	104,442.00	1.0	104,442.00
Transport & Logistics	3,000.00	1.0	3,000.00
Installation	6,000.00	1.0	6,000.00
PV mid / end clamp (each)	30.00	679.0	20,370.00
RR 10MM Copper Cable (100m coil)	2,500.00	2.0	5,000.00
Single core 4mm PV cable (100m coil)	2,500.00	4.0	10,000.00
63A DC MCB 2P	200.00	2.0	400.00
100A AC breaker	250.00	2.0	500.00
100A changeover	1,500.00	1.0	1,500.00
6-way breaker box	180.00	1.0	180.00
13-way breaker box	350.00	2.0	700.00

Item	Unit price (GHS)	Qty	Amount (GHS)
SPD / lightning arrester (500V class)	300.00	1.0	300.00
Miscellaneous (MC4, trunking, PVC, glands, fixings, lugs, ties)	1,500.00	1.0	1,500.00

Sub total: **1,034,792.00**

Tax (12.5%): **129,349.00**

Discount (0%): **0.00**

Option total **1,164,141.00**
(GHS):

SYSTEM SPECIFICATIONS & ENGINEERING DETAILS

DAILY ENERGY
REQUIREMENT

331.64 kWh/
day

EFFECTIVE DAILY
ENERGY

460.62 kWh/
day

After system losses

MONTHLY
GENERATION

13818.6
kWh

PEAK SUN HOURS

5.0 hrs/day

Location: N/A

PV ARRAY
CAPACITY

110.58 kW

194 × 570W panels

PANEL BRAND

Longi

570W model

INVERTER
CAPACITY

85.0 kW

Selected: 90.0kW
(for quote)

DC/AC RATIO

1.23

Optimized

BATTERY
CAPACITY

330.0 kWh

BACKUP
DURATION

24 hours

REQUIRED ROOF
AREA

605.3 m²

MOUNTING RAILS
(EST.)

90 sticks (18 ft)

≈ 491.4 m linear

SYSTEM
EFFICIENCY

72.0 %

Engineering Notes: This system is designed using industry-standard engineering factors, including system efficiency (accounting for inverter, wiring, and temperature), an optimized DC/AC ratio to prevent inverter clipping, and location-specific peak sun hours for accurate energy-generation calculations. All calculations include appropriate safety margins for reliable performance.

Mounting rail stick count is a **planning estimate** (18 ft segments, portrait grid); confirm with site layout and structural design.

Terms & Conditions

This quotation is valid for 30 days from the date of issue. All prices are subject to change without notice. Installation and project delivery terms will be discussed upon acceptance.

Payment Terms

Commercial off-grid hotel proposal — 30% deposit, 40% on equipment delivery, 30% on commissioning.

Notes

Client: Mr Collins · +233544036297 Site: Eastern Region, Ghana — 40-room hotel, restaurant & kitchen.
Objective: sufficient solar generation to operate off the national grid. Load per client brief: 16 ACs, 38 TVs, 100 bulbs, 36 fridges, 2 freezers, water pump, laundry & kitchen equipment, 55 fans.

Thank you for your business!